Course Code: CSE 3715	**Course Name:** Discrete Mathematics	
Credits: 2-1-0	**Contact hours per week:** 2 hours lecture & 1 hour discussion per week	
Batch: 2023, 5th Sem **Academic Year:** 2025-26	**Semester Duration:** 29 July 2025 to 28 Nov 2025	
Instructor-in-charge: Dr. Aradhana Narang Email - aradhana.narang@bmu.edu.in Phone -7550160204 Office –22, Fourth Floor, E2 Building Dr. Luckshay Batra Email – luckshay.batra@bmu.edu.in Phone - 9910931128 Office –88, Fourth Floor, E2 Building Dr. Satpal Singh Email – satpal.singh1@bmu.edu.in Phone -8949204178 Office - Fourth Floor, E2 Building	**Course Coordinator –** Dr. Aradhana Narang Email - aradhana.narang@bmu.edu.in Phone -7550160204 Office –22, Fourth Floor, E2 Building	BML MUNJAL UNIVERSITY FROM HERE TO THE WORLD

Aim of the course: This course is aimed at helping students understand and develop a background in theoretical computer science. The main objective of the course is to teach the basics of discrete mathematics and apply and correlate them in other computer science courses.

Course Overview:

This is an introductory undergraduate course on Discrete Mathematics. The focus will be on the discrete mathematical structures with application to computer science. The topics to be covered include basic set theory, relations and functions, number theory, principle of mathematical induction, propositional logic, proof techniques, combinatorics and algebraic structures. Salient topics of the course are listed below:

Week-Wise Schedule:

Note: *Weekly course schedule includes 02 lecture sessions and 01 discussion*

Week 1-3: **Sets, Relation and Functions**
Operations and Laws of Sets, Cartesian Products, Binary Relation, Partial Ordering Relation, Equivalence Relation, Image of a Set, Bijective functions, Inverse and Composite Function, Size of a Set, Finite and infinite Sets, Countable and uncountable Sets, The Power Set theorem, Schroeder-Bernstein theorem.

Week 4: **Principles of Mathematical Induction**
- The Well-Ordering Principle
- Weak and strong induction
- Recursive Definition
- Recurrence Relations

Week 5-7: **Number Theory and Cryptography**
- The Division algorithm, The Greatest Common Divisor
- Prime Numbers, The Fundamental Theorem of Arithmetic, Pseudo Primes
- Modulo arithmetic, Linear congruences, Chinese remainder theorem.
- Application of congruence and applications to cryptography.

Week 8-9: **Propositional Logic**
- Basic Connectives and Truth Tables
- Logical Equivalence

- The Laws of Logic, Logical Implication
- Rules of Inference, Validity of Arguments
- The use of Quantifiers, Predicate Logic
- Arguments in Predicate Logic

Week 10: **Proof Techniques**

- Some Terminology, Proof Methods and Strategies
- Direct Proof, Proof by Contradiction, Proof by Contraposition, Proof of Necessity and Sufficiency.

Week 11: **Combinatorics**

- Counting Principles, Functions and Counting
- Pigeonhole Principle
- Permutations and Combinations, Combinatorial Arguments

Week 12-16: **Algebraic Structures**

-Algebraic Structures with one binary operation: Semi Groups, Monoids, Groups, Cyclic Groups, Permutation Groups, Normal Subgroups.

-Algebraic Structures with two Binary Operation: Rings, Integral Domain, and Fields.

-Boolean algebra and Boolean Ring, Identities of Boolean algebra, Duality, Representation of Boolean Function, Disjunctive and Conjunctive Normal Form.

Course Outcomes:

COs	Course Outcome Statement	Mapped Component	Associated activities/Assessment
CO1	Understand basic set theory, functions and relations, principle of mathematical induction, recurrence relations, basic and advanced counting techniques, number theoretic techniques, cryptography, predicate and propositional logic, rules of inference, formal and informal methods of proofs, algebraic structures.	Skill Development	-Quiz -Assignment -End Term Exam
CO2	Apply concept of relations to find matrix of a relation, equivalences classes and partitions of a set, concepts of digraph and poset, concepts of logic to translate a given sentence in terms of predicates, quantifiers, and logical connectives, rules of inference and methods of direct, indirect proofs for proving results. Apply the theory of numbers to solve congruence related problems and cryptography. Application of counting techniques, principle of	Skill Development	-Quiz -Assignment -End Term Exam -Project (Use of C/C++/Python)

	mathematical induction and recurrence relations, algebraic structures, Boolean algebra		
CO3	Calculate truth value of a sentence, Analyze logical equivalences and validity of an argument using predicate logic. Find and analyze asymptotic bounds on primes using number theoretic techniques.	Skill Development	-Quiz -Assignment -End Term Exam -Project (Use of C/C++/Python)
CO4	Model real world computer science problems and find their solutions using computer algorithms, programs and write a project to summarize the findings.	Skill Development, Employability	-Project (Use of C/C++/Python)

CO-PO Mapping

PO and PSO →	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO2**	**PSO3**	**PSO4**
↓CO Mapping || Course Mapping →																
CO1	1															
CO2	3	2		2	2											
CO3	3	2		2	2											
CO4	3	2		2	2				2	2						

Note: For details about PO, PSO, and CO-PO mapping, refer to OBE document for B. Tech. – CSE program.

1- **Low Correlation; 2- Moderate Correlation; 3- Substantial Correlation**

Course Competencies (Course Outcomes further elaborated) and Instruction schedule:

	Competency	**CO**	**CL**	**No of Contact hours**

C1	**Sets, Relation and Function** - Operations and laws of sets - Binary Relation, Partial Ordering Relation, Equivalence Relation - Size of a Set, Finite & infinite Sets, Countable & uncountable Sets - The Power Set theorem, Schroeder-Bernstein theorem.	CO1, CO2	U AP	6+3
C2	**Principles of Mathematical Induction** - The well-ordering principle - Normal and strong induction - Recursive definition - Recurrence Relations	CO1, CO2, CO4	U AP EV	2+1
C3	**Number Theory and Cryptography** - The Division algorithm, The Greatest Common Divisor - Prime Numbers, The Fundamental Theorem of Arithmetic, Pseudoprimes - Modulo arithmetic, Linear congruences, Chinese remainder theorem. - Application of congruence and applications to cryptography.	CO1, CO2, CO3, CO4	U AP AN EV	6+3
C4	**Propositional Logic** - Basic Connectives and Truth Tables - Logical Equivalence, The Laws of Logic, Logical Implication - Rules of Inference, Validity of Arguments - The use of Quantifiers, - Predicate Logic, Arguments in Predicate Logic	CO1, CO2, CO3, CO4	U AP AN EV	4+2
C5	**Proof Techniques** - Some Terminology, Proof Methods and Strategies - Direct Proof, Proof by Contradiction, Proof by Contraposition - Proof of Necessity and Sufficiency.	CO1, CO2	U AP	2+1
C6	**Combinatorics** - Counting Principles, Functions and Counting - Pigeonhole Principle - Permutations and Combinations, Combinatorial Arguments.	CO1, CO2, CO4	U AP EV	2+1
C7	**Algebraic Structures:** -Algebraic Structures with one binary operation: Semi Groups, Monoids, Groups, Cyclic Groups, Permutation Groups, Normal Subgroups. -Algebraic Structures with two Binary Operation: Rings, Integral Domain, and Fields. -Boolean algebra and Boolean Ring, Identities of Boolean algebra, Duality, Representation of Boolean Function, Disjunctive and Conjunctive Normal Form.	CO1, CO2, CO4	U AP EV	10+5

Total contact hours=48

Learning Resources:

Textbooks:

1. Kenneth Rosen, Discrete Mathematics and Its Applications, 7th Ed, McGraw Hill Publishing Co., 2012.

Reference Books:

1. Jean Paul Trembley, R. Manohar, Discrete Mathematical Structures with Application to Computer Science, McGraw-Hill.
2. C. Liu and D. Mohapatra, Elements of Discrete Mathematics: A Computer Oriented Approach, Tata McGraw-Hill Education.
3. R.P. Grimaldi, "Discrete and Combinatorial Mathematics", Addison Wesley.
4. B. Kolman, R.C. Busby, and S.C. Ross, "Discrete Mathematical Structures", PHI Learning Private Limited, India

E-Learning (and other) Resources:

1. https://www.cs.ox.ac.uk/teaching/courses/discretemaths/
2. https://onlinecourses.nptel.ac.in/noc20_cs37/preview
3. https://www.coursera.org/specializations/discrete-mathematics
4. https://www.classcentral.com/course/swayam-discrete-mathematics-12929
5. https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-042j-mathematics-for-computer-science-fall-2010/

Assessment Pattern:

Assessment Pattern: The final grade will be based on the marks obtained in various assessment components defined in the assessment table below. Relative grading method defined in the academic regulations of the university will be followed to grade the students. Student must secure minimum 40% of marks after completing all the assessments to become eligible for grading.

Evaluation Component	**Weightage (%)**	**Evaluation Schedule**	**Rubrics/Remarks**
Quiz	20 (2 quizzes each of weightage 10%)	21 August 2025 10 October 2025	Each quiz is 25 minutes long closed book assessment component. Understanding of Concepts (40%) • Proper understanding of concepts of discrete structure. • Validation of students' understanding through timed quiz. Problem Solving (60%) • Applying the knowledge to solve various engineering problems • Logical ability to problem-solving skill in a timely

			manner.
Assignment	20 (2 assignments each of weightage 10%)	15 September 2025 28 October 2025	Two open book assignments each of weightage 10%. Understanding of Concepts (20%) • Detailed understanding of various concepts and terminologies of discrete structure. Timeliness (20%) • Assignment must be submitted on or before the due date and time. Problem Solving (60%) • Applying mathematical knowledge to solve various problems arising in different engineering domains. • Developing problem-solving skills.
Project	20	November 19, 2025 (Presentation: during Nov 26-28, 2025 in the class timings)	Understanding of Concepts (10%) • Comprehensive understanding of the concepts taught throughout the course. • Understanding of various types of applications. Timeliness (10%) • Report must be submitted on or before the due date and time. Report Writing and Problem Solving (30%)

			• Developing in-depth understanding to prepare a detailed report of the findings. • Applying the knowledge from the course to solve real life applications. • Demonstration of coding skills. • Visualization of the problems using softwares. Presentation and Viva (50%) • Developing peer-learning and collaboration skills. • Developing presentation skills to present the work in a group. • Demonstration of individual knowledge through viva.
End Semester	40	As per University Calendar	Comprehensive Knowledge (40%) • Demonstrates understanding of entire course material • Ability to apply concepts to solve complex problems Practical Application (40%) • Application of theoretical knowledge in practical scenarios. Critical Thinking (20%)

			• Analytical skills and innovative solutions to handle real-life problems. • Ability to analyze the efficiency of various available methods to apply the suitable one for time optimization and improve credibility.

Experiential Learning Components:

Hands-on projects on applications of Discrete Mathematics in C/C++/Python will be given to the students in which they will work in group on applications of counting techniques to solve problems in computer science and related areas. Application of recurrence relations to generate random numbers, OTP, Application of number theoretic techniques to validate a given ISBN number, Bank Cheque number, Aadhar/Social security number. Applications of propositional logic to computer engineering problems. Application of Boolean Algebra to Logic Gates etc. This entire set of learning will be assessed through the evaluation component of 20% weightage.

Student Responsibilities:

- Attend lectures to obtain all the course material that you are responsible for.
- Check announcements at Maitri/Google Classroom on a regular basis.
- Submit assignments/projects on time.

Regularly, check your marks on the Maitri and inform the instructor in-charge in case of any issues.

Attendance Policy: Students are expected to attend the classes regularly. Failure to attend the classes regularly and adhere to the expected attendance percentage may result in a reduction of the grade or awarding 'R' grade as per the University's grading policy.

Recourse examination policy: As per institute policy.

Late assignment submission policy: 5% of the total marks would be deducted for late submission.

Make-up examination/work policy: As per University policies.

Behavior expectations: No mobile phones and other distractive elements and gadgets in the class.

Other Policies or Procedures:

a) Academic Honesty

Students must abide by BMU's academic honesty policy. For example, an excerpt is provided for guidance as follows:

"Students who compromise the integrity of the classroom are subject to disciplinary action on the part of

the University. Violations of classroom standard include:
1. Cheating in any form, whether in formal examinations or elsewhere
2. Plagiarism, using the work of others as your own without assigning proper credit to the source.
3. Misrepresentation of any work done in the classroom or in preparation for class.
4. Disruptive behaviour in a course of study or abusiveness toward faculty or fellow students."

b) Teaching Improvement

Please feel free to make suggestions to improve the content of the class and my instruction skills. You can share these suggestions directly to me and/or via email. You can also leave your comments anonymously by sending online messages.